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Deterministic interference, exact branch counts, and quantum measurement from the fold

Maria Smith
Ernos Labs

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EXACT BRANCH CENSUS
8 branches x 1/8 = the One

Quantum probability is a discrete preimage count; the fold contains no primitive die.

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Abstract

Smithian Fold Theory reconstructs the quantum measurement programme from one machine-checked self-proven theorem, zero axioms and zero fitted parameters. The resulting object is exact and deterministic. At observation depth three it generates eight branches, gives each branch the indivisible weight 1/8, and proves that the complete branch count partitions the One. The same derivation produces a discrete stationary spectrum with spacing 1/8, ground share 1/16, exact phase return, and a deterministic optics relation in which maximal mismatch 1/2 is self-antipodal, opposed shares cancel, aligned halves complete to the One, and dark plus bright shares conserve the whole.

The conclusion is direct: quantum probability is a ratio of discrete preimage counts, not a primitive continuum lottery. A single outcome can be determined while remaining unavailable to a state that cannot invert the fold that produced it. The apparent die roll is therefore structural self-opacity, not an uncaused event. The theory also derives the two exchange classes - symmetric coincidence and antisymmetric distinction - from a binary fibre and proves that they exhaust it.

This is not an opaque model whose reliable output substitutes for an explanation. Every numerical result is generated by named relations, expressed in exact arithmetic, compiled into independent C certificates and reproducible from the public source. Empirical validity means that the construction, the execution and the measured consequence remain connected by a trace that can be inspected and rerun. The paper states that construction and its experimental programme without inserting statistical prediction as a higher authority than mathematical proof.

Central result

Layer	Exact result
Foundation	One self-proven theorem; zero axioms; zero fitted parameters
Observation depth	3, the colour count
Branch count	$2^3 = 8$
Atomic branch weight	1/8
Completeness	$8 * (1/8) = 1$
Stationary spacing	1/8 at depth three
Ground share	1/16; one fold gives 1/8
Maximal mismatch	1/2, exactly self-antipodal
Bright completion	$1/2 + 1/2 = 1$
Optics conservation	dark share plus bright share equals the One
Exchange census	one symmetric and one antisymmetric class; no third class

1. The empirical question

The double-slit experiment combines two facts: detections arrive as individual events, while the complete record forms bright and dark interference bands. Standard probability language correctly organizes the observed frequencies, but a useful formula does not by itself determine what the probability is physically.

SFT asks a sharper question: can the weights, stationary modes and interference relation be generated as exact consequences of one finite operation? Its answer is yes. The engine contains no random-number source, probability parameter or fitted amplitude. It counts preimages, walks exact fold orbits and checks the closure of the complete object.

2. Foundation: the One and its non-invertible fold

The corpus begins with a machine-checked self-proven theorem rather than an assumed axiom. The theorem forces the One and the fold. The fold doubles a magnitude and casts out completed Ones. Its binary fibre is two-to-one: two preimages can share one image.

That non-invertibility matters. Forward evolution remains exact, while inversion is not available from the image alone. A state can therefore be fully determined by its route without carrying enough information to reconstruct which preimage supplied it. Determination and foreknowledge are not the same operation.

3. Probability as a preimage census

The observation depth is the forced colour count:

$$k = 3.$$

The branch count is therefore

$$B = 2^k = 2^3 = 8,$$

and the atomic branch share is

$$w = 1 / 2^k = 1/8.$$

The engine separately verifies the completeness relation

$$B * w = 8 * (1/8) = 1.$$

The ratio attached to an outcome is consequently a discrete census of its preimages within the whole. It is exact because the numerator and denominator are counts. Nothing is sampled from an assumed continuum, and there is no trained parameter whose numerical value hides the construction.

4. Deterministic interference

The wave-optics certificate derives the extrema of the interference relation:

1. maximal mismatch is $1/2$;
2. the mismatch is self-antipodal under the fold;
3. aligned half-shares complete exactly to the One;
4. the dark and bright shares together conserve the One.

This gives a constructive reading of the bands. Opposed routes cancel at the self-antipodal point; aligned routes complete. Bright and dark are generated relations of the same conserved whole, not arbitrary labels attached after observation.

5. The stationary comb

At depth three the stationary-state engine gives the exact level spacing

$$\Delta = 1/8$$

and the ground share

$$E_0 = 1/16.$$

One fold advances the ground share to the spacing, two ground shares make one step, every adjacent gap is one step, and the phase returns after a complete cycle. For the second excited index the certificate returns

$$E(2) = 1/16 + 2/8 = 5/16.$$

The interference record is therefore supported on a counted stationary structure rather than an unconstructed continuum of possibilities.

6. Why the single event can appear random

The engine's branch census determines the complete relation, but a branch image does not encode which of its two preimages supplied it. The apparent unpredictability of the single event is the operational consequence of that non-invertibility. It does not require nature to choose without cause.

This is the distinction the paper establishes:

- **determinism** concerns whether the forward relation is fixed;
- **self-knowledge** concerns whether a state can invert the relation from inside it;
- **experimental frequency** counts how often a registered outcome is reached across the branch census.

The three are related but not interchangeable.

7. The exchange census

The quantum-statistics engine starts from the forced binary fibre of size two. Exchange then admits two and only two structural cases:

- coincident images are fixed by exchange and therefore symmetric;
- distinct images reverse orientation and therefore antisymmetric.

The same binary census gives the fermion occupancy cap: a third distinct member has no preimage in the fibre. The machine certificate checks Bose symmetry, Fermi antisymmetry, the Pauli cap and exhaustion of the two cases.

8. Nonlocality without a signalling mechanism

The fold's referent is the One, so composite correlation is constructed globally rather than assembled from independent local variables. Bell-type evidence against local hidden variables does not impose randomness on a global deterministic construction. The companion paper *Entanglement Without Spookiness* gives the exact shared-origin and no-extra-channel derivations.

9. What counts as empirical evidence here

Opaque reliability is not the same evidential object as transparent empirical proof. A system can produce accurate outputs while concealing the relations that produced them. SFT requires the stronger chain:

1. the theorem forces the primitive object;
2. each additional relation is derived or the engine halts;
3. exact source executes the relation;
4. generated certificates independently reproduce it;
5. measurement compares the declared consequence with nature.

Mathematical proof establishes necessity inside the formal construction. Machine execution establishes that the published implementation realizes that construction. Experiment establishes whether the construction describes the physical object. All three are empirical parts of the programme; none is replaced by a black-box score.

10. Reproduction

From the public repository root:

```
./verify/test_measurement_problem
./verify/test_measurement_branch_weight
./verify/test_quantum_stationary_states
./verify/test_quantum_statistics
./verify/test_wave_optics
```

The current focused run returns every registered check as ok. The authoritative corpus release also supplies the source-to-generated-C identity audit and the complete ordered derivation in `OneFoldMaster.md`.

11. Empirical programme

The construction supports direct experimental work rather than a theoretical wall:

- resolve whether allowed and cancelled interference structure is better represented by the derived discrete comb than by an ontic continuum;
- test branch-frequency data against exact integer preimage censuses;
- test stationary-level relations at increasing fold depth;
- design single-event protocols that distinguish structural self-opacity from an assumed primitive random source;
- cross-check the two-class exchange census wherever composite statistics can be cleanly isolated.

Because the relations are exact and parameter-free, the programme cannot rescue a mismatch by refitting a weight. The next stage is therefore additional measurement, not a retreat from the derivation.

Conclusion

The fold does not throw dice. It generates eight exact branches of weight $1/8$, closes them to the One, constructs a stationary comb, produces deterministic bright-dark interference and exhausts quantum exchange with two structural classes. Probability is the normalized count of preimages; apparent single-shot randomness is the self-opacity of a non-invertible relation. The complete result is public, exact and machine-checkable.

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